UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2026-2027

Imaging of the Mediastinum

Department of General Surgery/ Diagnostic Radiology

Stage 4 • Academic Year 2026–2027

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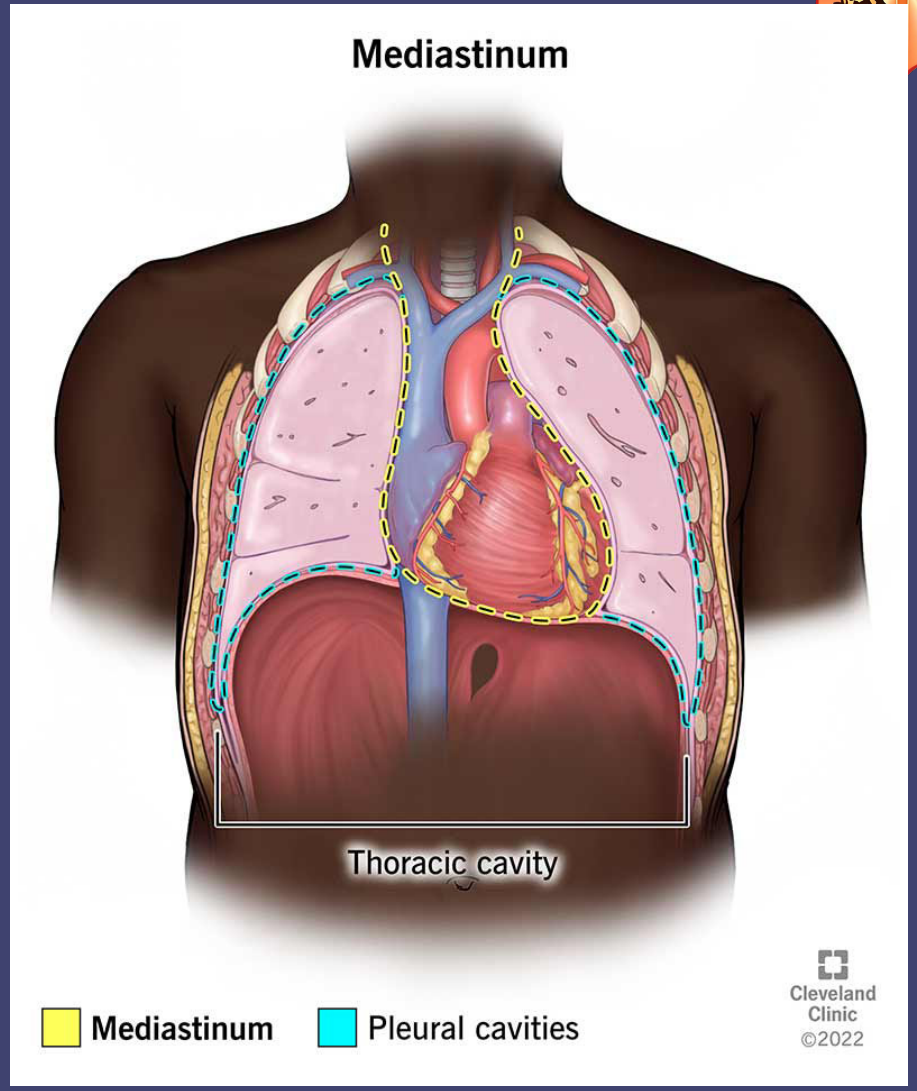
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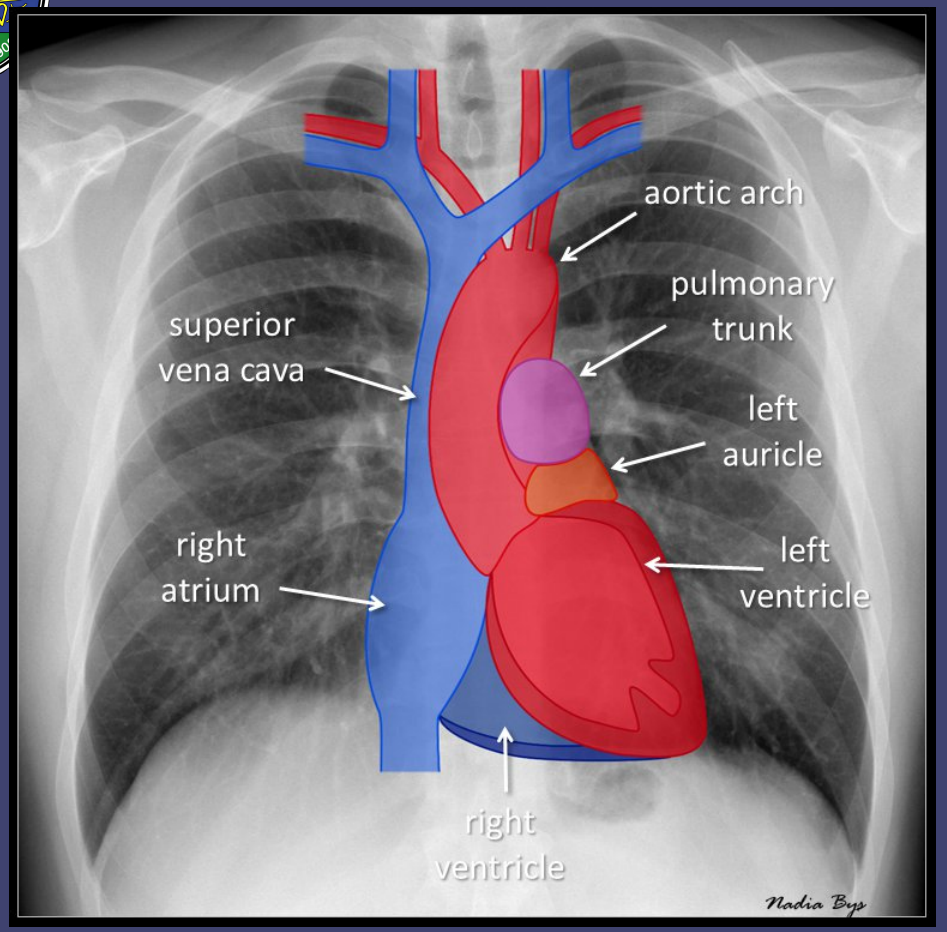
Learning Objectives

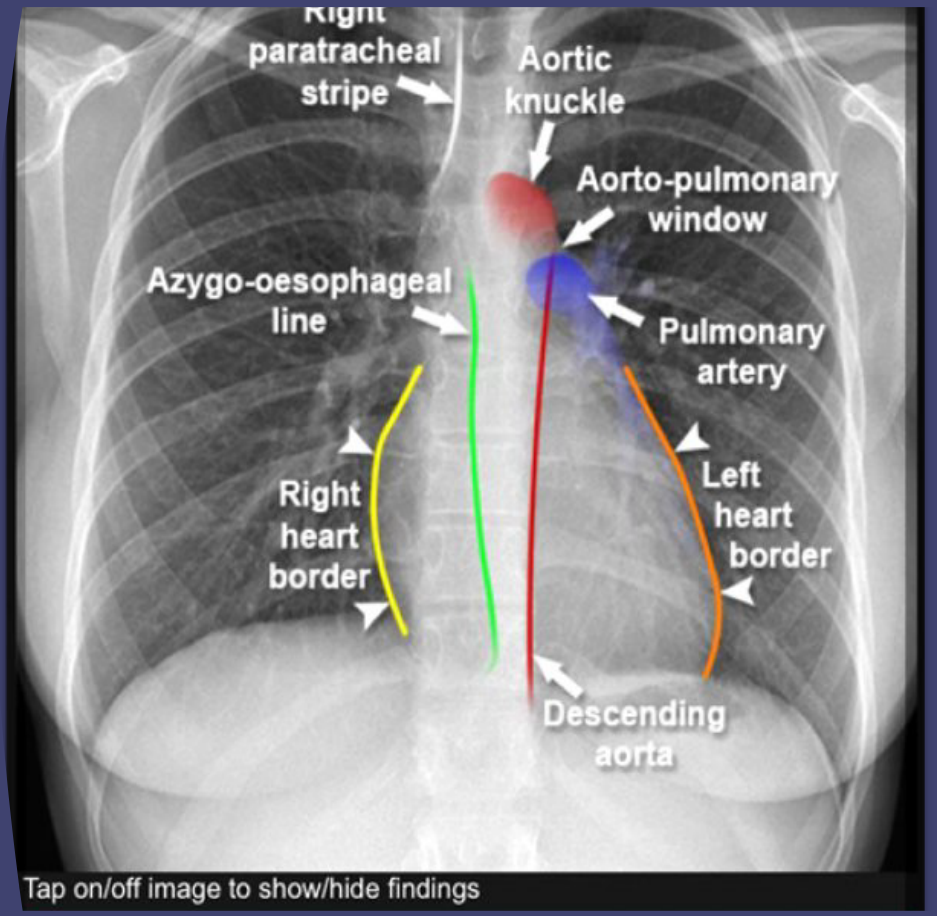
- Understand regional anatomy and relvant pathology related to each mediastinal compartment
- Identify specific radiological features to guide the diagnosis for mediastinal and hilar pathologies
- Gain insight into the optimal use of chest radiography, computed tomography (CT) for characterizing diseases.
- Identify the appropriate location for common devices/tubes (seen on CXR) and the complication related to misplacement

Anatomy

- The mediastinum consist of multiple midline structures that separates both lungs.
- It includes the heart, major vessels (pulmonary and aorta), trachea and bronchi, esophagus, lymph nodes, nerves and paravertebral region.
- Its divided into superior mediastinum, anterior mediastinum, middle mediastinum and posterior mediastinum.
- Each compartment contain different structures and different range of pathologies.





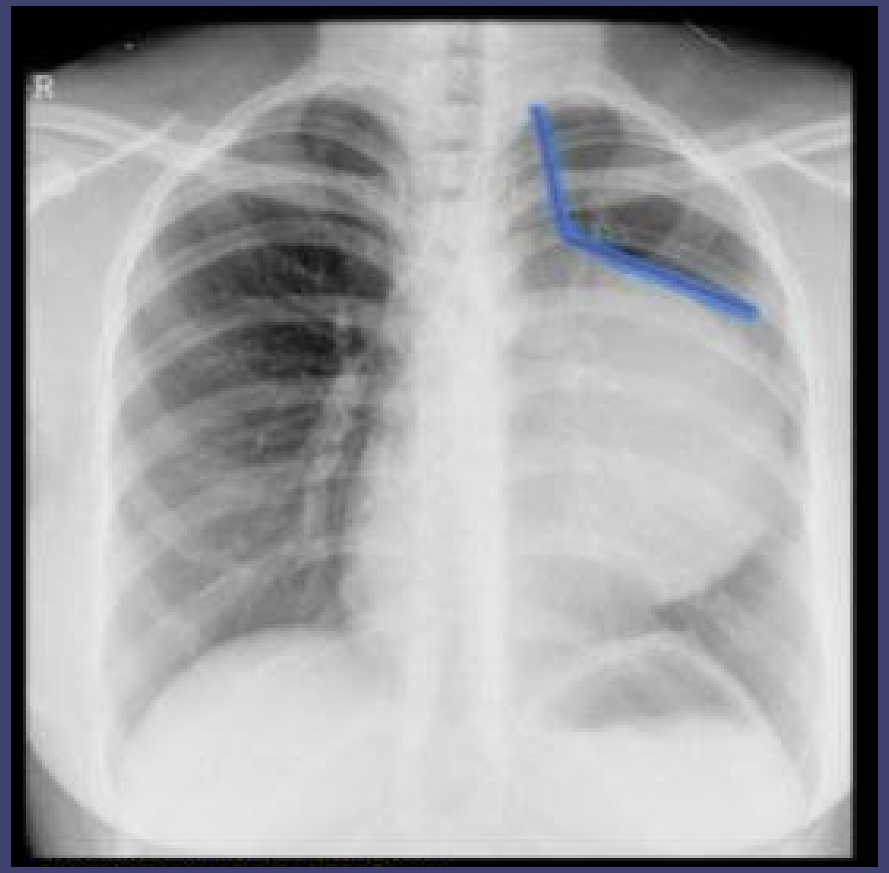


Normal mediastinal structures on CXR

Characteristics of mediastinal mass on CXR

The major features are :

- **Widening** of mediastinum.
- Change in contour/position of mediastinal structures (for e.g. deviation of the trachea in retrosternal goiter).
- **Mass that forms obtuse angle** (greater than 90 degrees) with the lung edge.

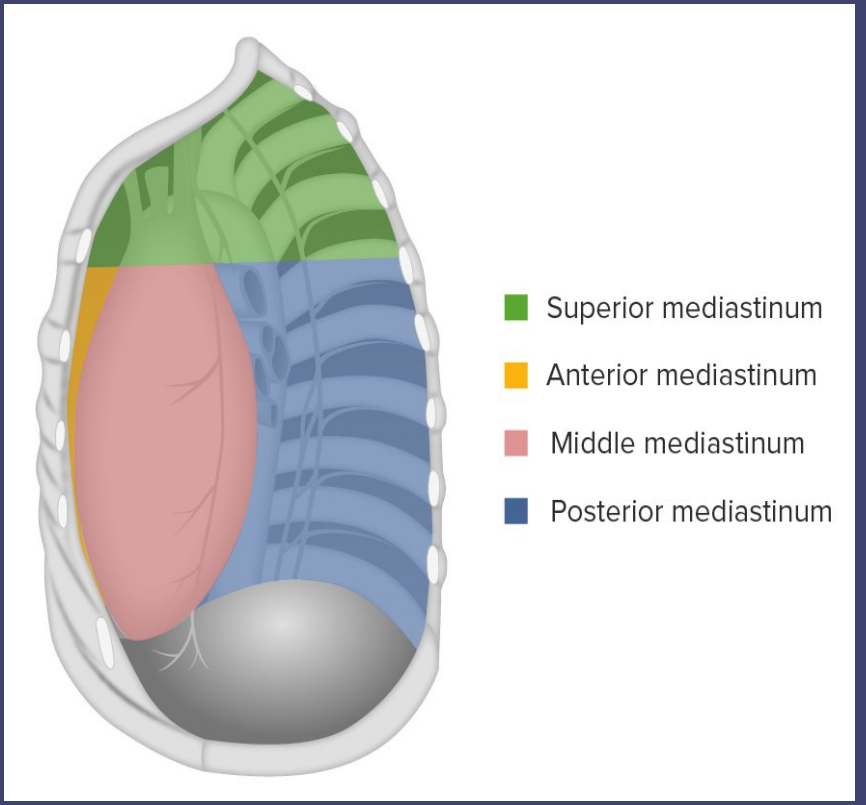


Imaging of the Mediastinum

- **A- CXR :**

- Usually performed as PA and LATERAL VIEWs, the lateral view localizes the mass within the mediastinum (Ant/Mid/Post)
- Can detect the abnormality but usually suboptimal for characterization.

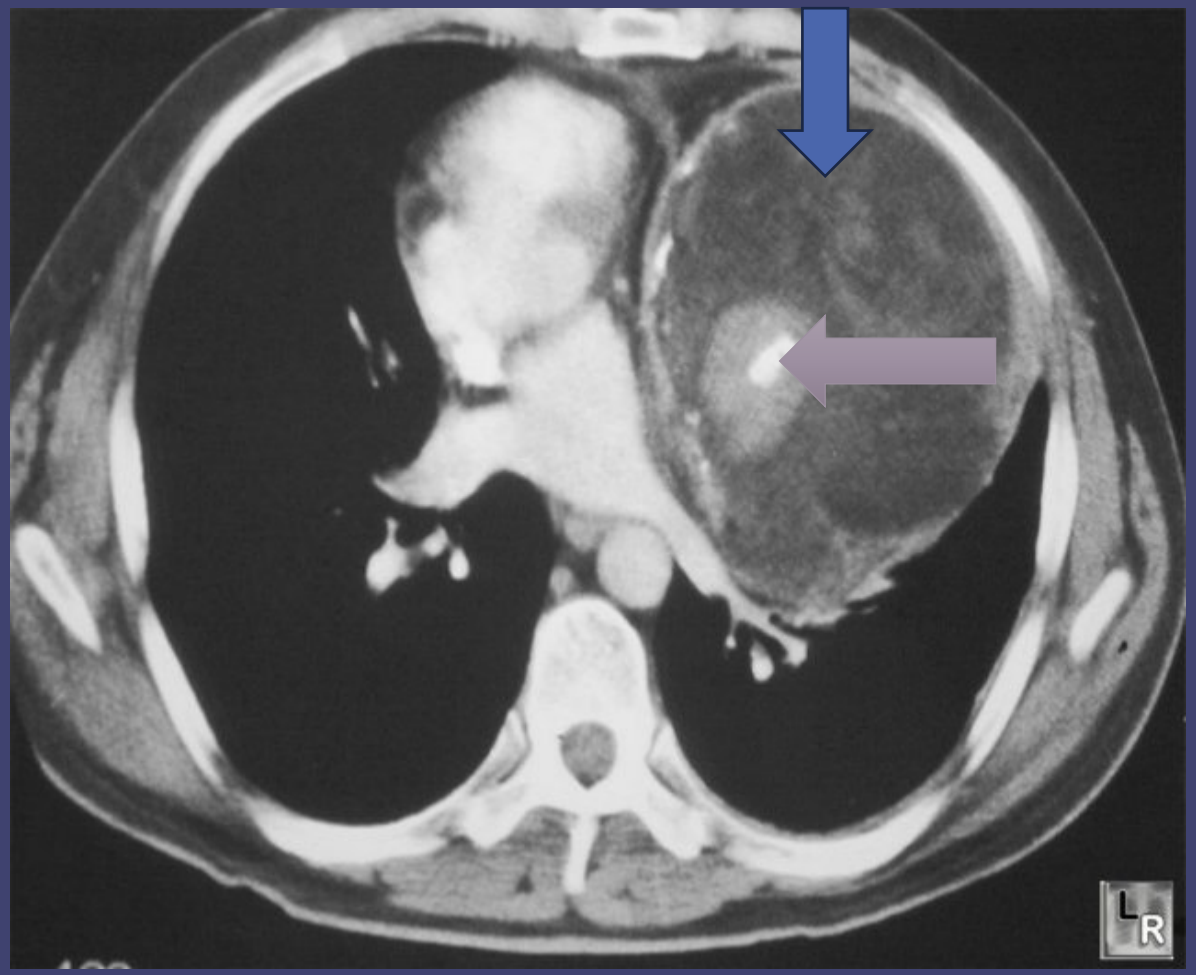
- **So : when a mediastinal lesion is seen on CXR it should be further characterized with CT-scan.**



Advantages of CT scan

- Definitive characterization of mediastinal mass :
- Location
- Contents (fat/calcium/soft tissue) as each mass contains different type of tissue which help in their characterization.

- Operative planning : extension/invasion of structures particularly major vessels.
- Staging of malignant lesions.
- Definitive dx of vascular emergencies which require urgent intervention (ruptured aortic aneurysm).



CT is used for assessment as the following

- Location: Ant mediastinum
- Content: Fat (blue) and calcium (purple)
- Operative planning : its in contact with LT atrium and pulmonary veins
- Non invasive

Divisions of the Mediastinum

- The mediastinum is divided into different compartments including

1. Anterior Mediastinum
2. Middle Mediastinum
3. Posterior Mediastinum

- Each compartment contain different anatomical structures so accurate localization is required for appropriate evaluation; for example :

The thymus is normally present in the anterior mediastinum so thymic tumors are only seen as anterior mediastinal mass (should not be included in the DDX of middle or posterior mediastinal lesions).

- Some pathologies like lymphadenopathy and aortic aneurysm can be seen in all three mediastinal compartments.

Anterior mediastinal mass

The DDX of anterior mediastinal mass includes 5 lesions all of which begin with the letter T.

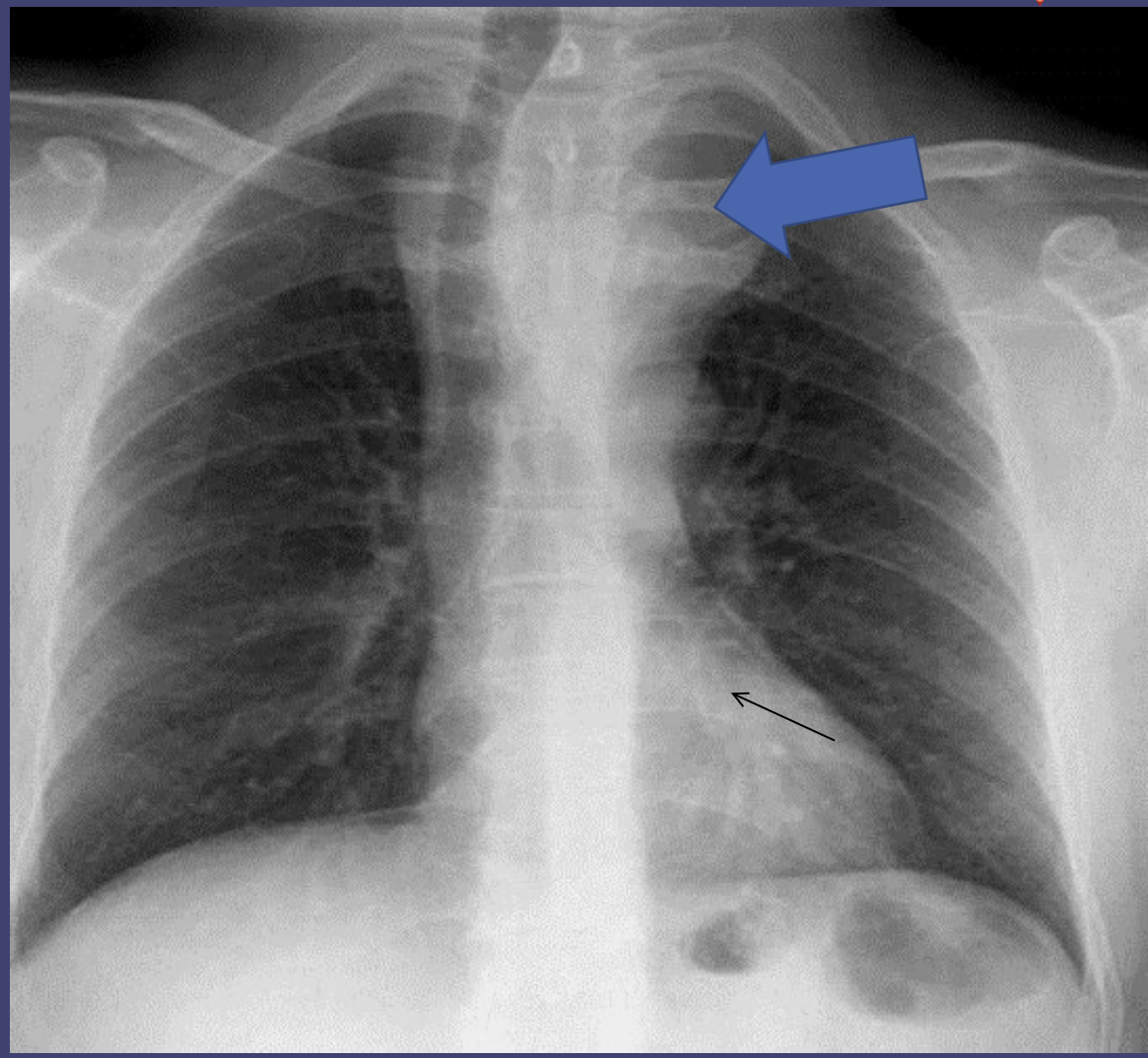
- Thyroid retrosternal Goiter
- Thoracic aortic aneurysm
- Terrible lymphoma
- Thymus tumors
- Teratoma



•Thyroid

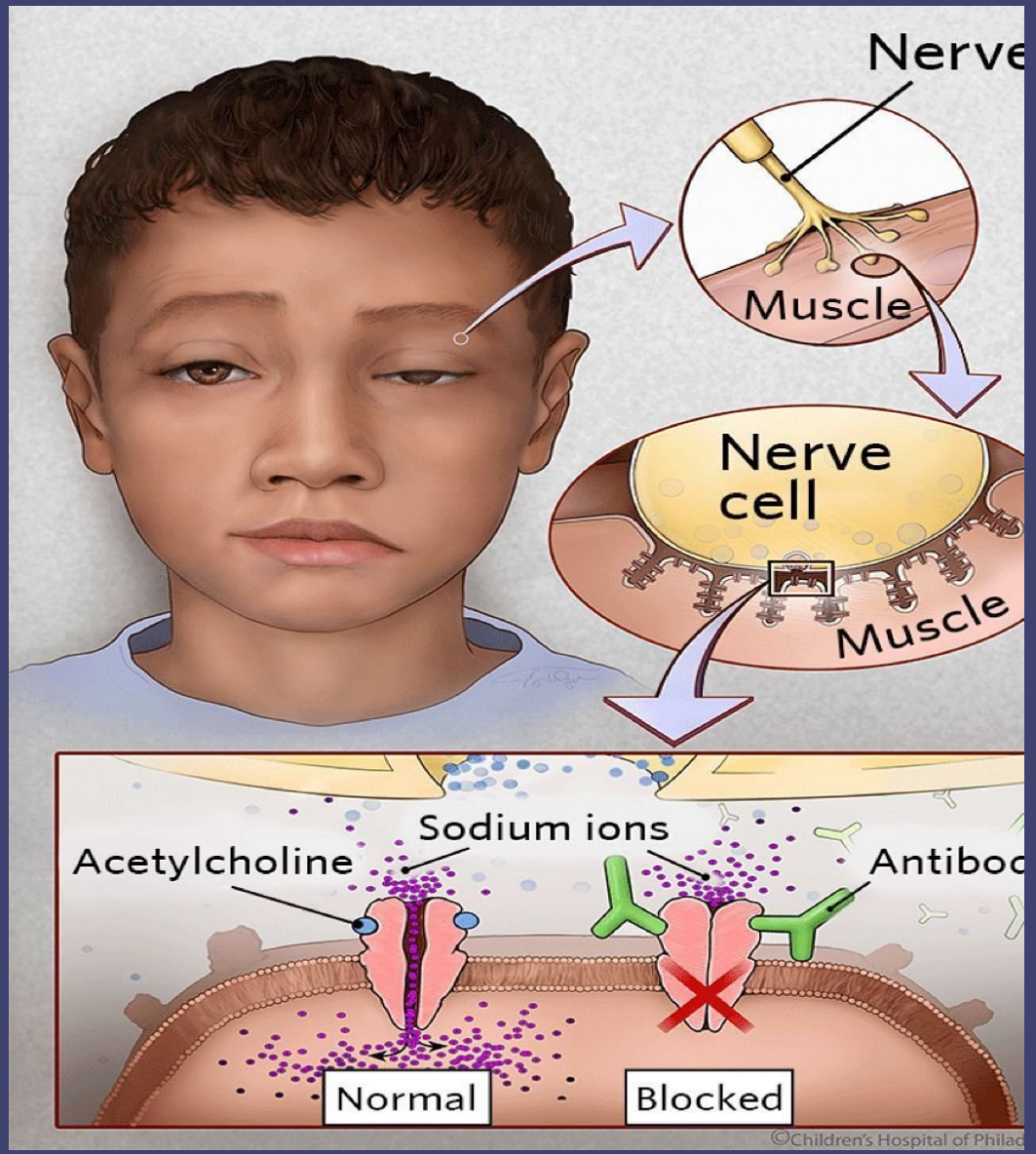
- Retrosternal goiter can present as anterior mediastinal mass.
- On x ray : it pushes the trachea to the other side.
- CT can definitely show the extent of retrosternal extension and helps in surgical planning.

• Note the enlarged thyroid gland on CXR



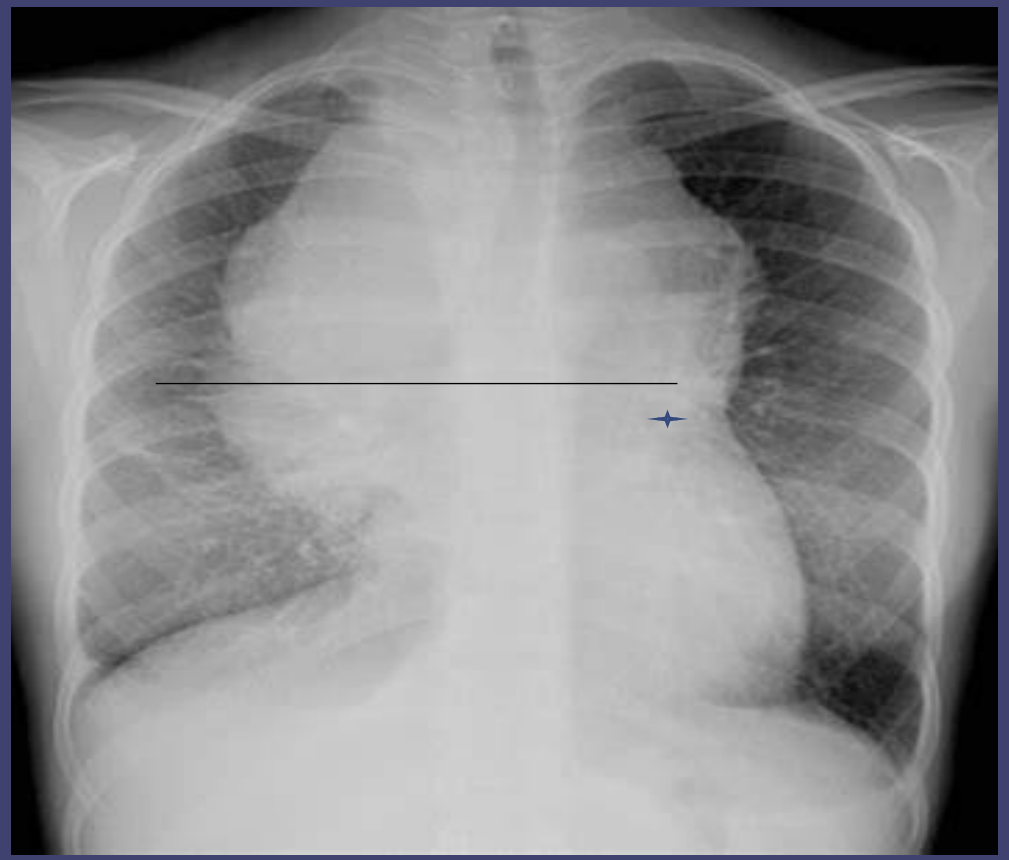
Thymic tumors

- Thymoma is a tumor of the thymus gland that presents as anterior mediastinal mass.
- It is associated with myasthenia gravis, a neuromuscular disorder that causes muscle weakness.
- Identification of the mass is essential for Mx.

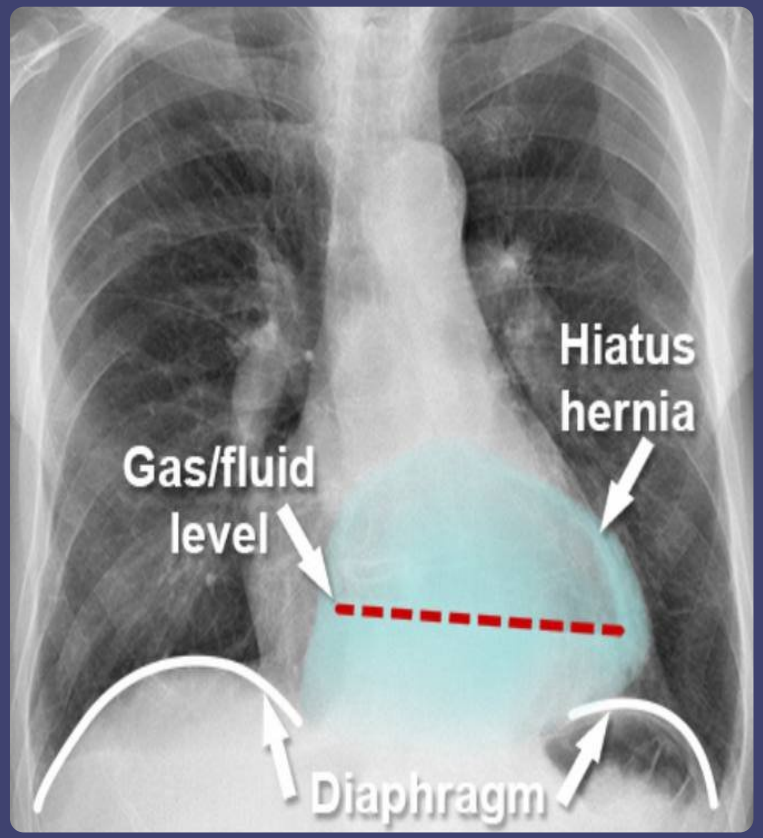


•Terrible Lymphoma

- Lymphoma is an example of anterior mediastinal mass
- It can involve all three compartments of the mediastinum (anterior,middle,posterior).
- Can be associated with hilar lymphadenopathy /enlarged nodes in other parts of the body.



- Hiatal Hernia is a frequent example of posterior mediastinal pathology on CXR





Aortic aneurysm

- **Ruptured thoracic aortic aneurysm is a top emergency** which requires early detection and surgical intervention.

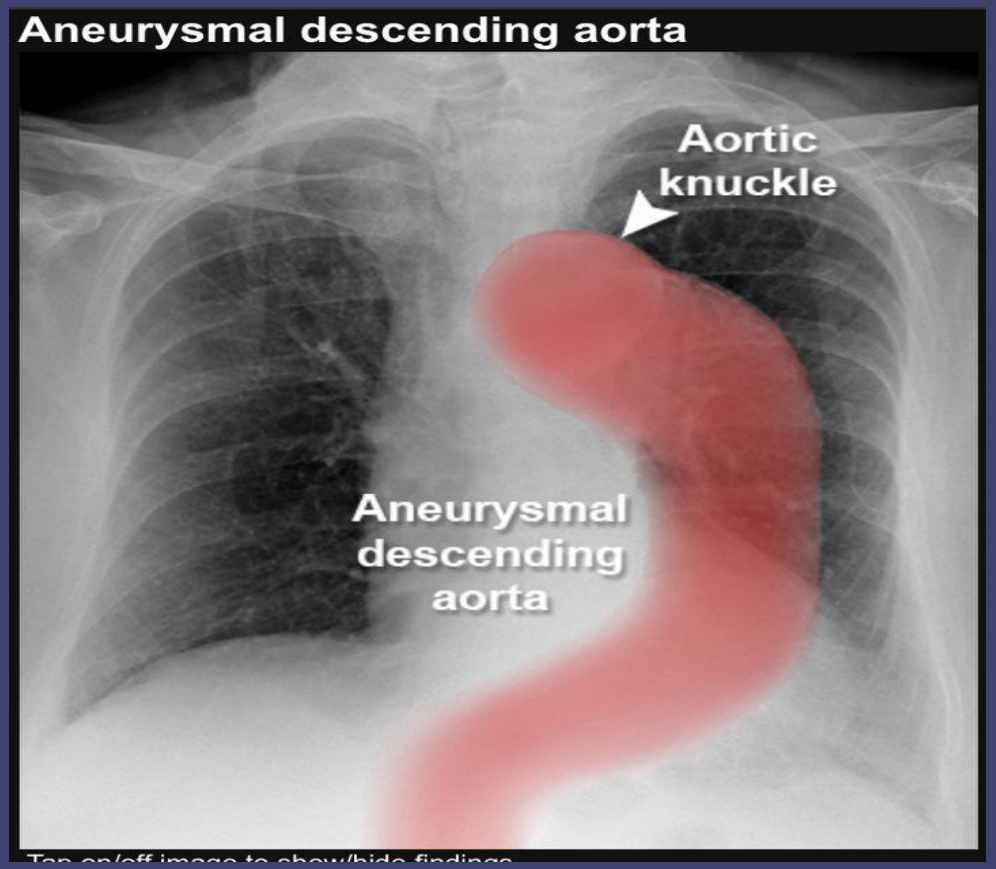
- It can be seen on CXR as a mediastinal widening. But definitive evaluation can not be made.

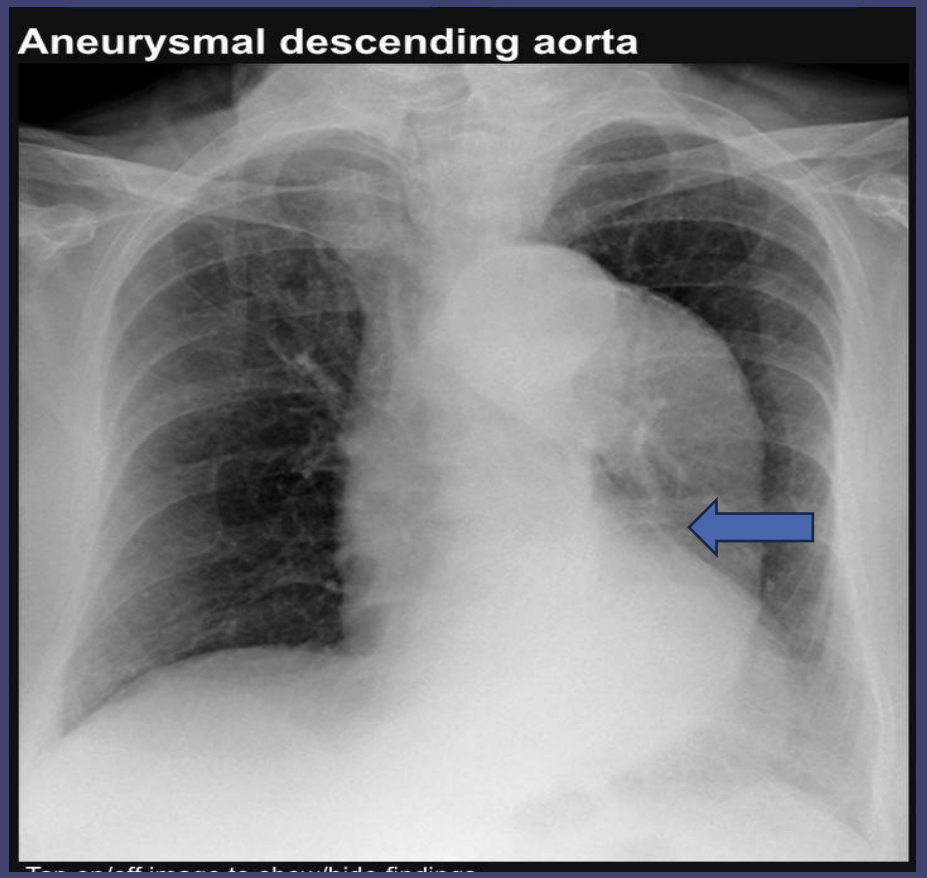
Be aware that normal CXR does not exclude Aortic dissection and CT with contrast is always required for assessment.

- In acute aortic syndrome, CT scan with contrast should be performed to confirm the dx and detect other complications.

- The main advantages of CT scan

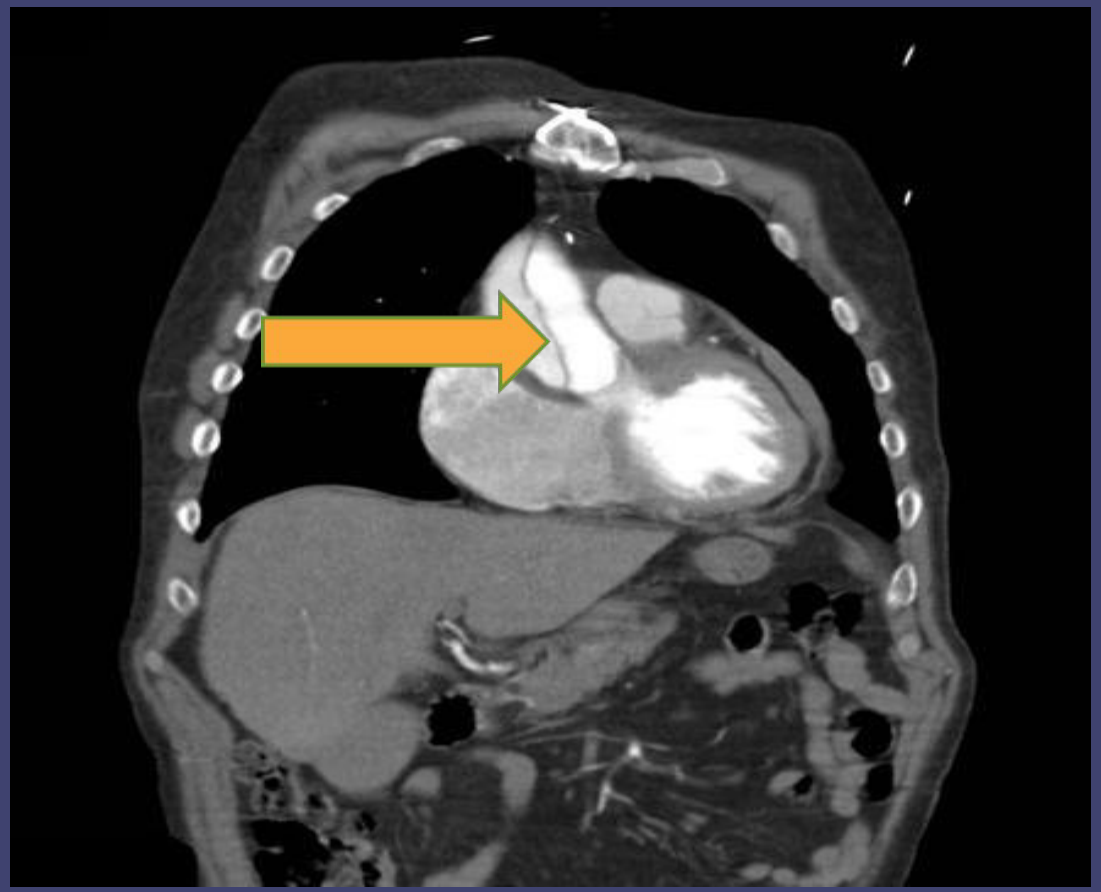
1. Classification of aneurysm.
2. Extension to abdominal aorta
3. Involvement of other branches like carotids/coronaries,
4. Features of rupture like hemothorax.





Aortic aneurysm is usually seen on CXR however CT is required for definitive assessment

CT chest with contrast showing aortic dissection (CXR can be entirely normal).

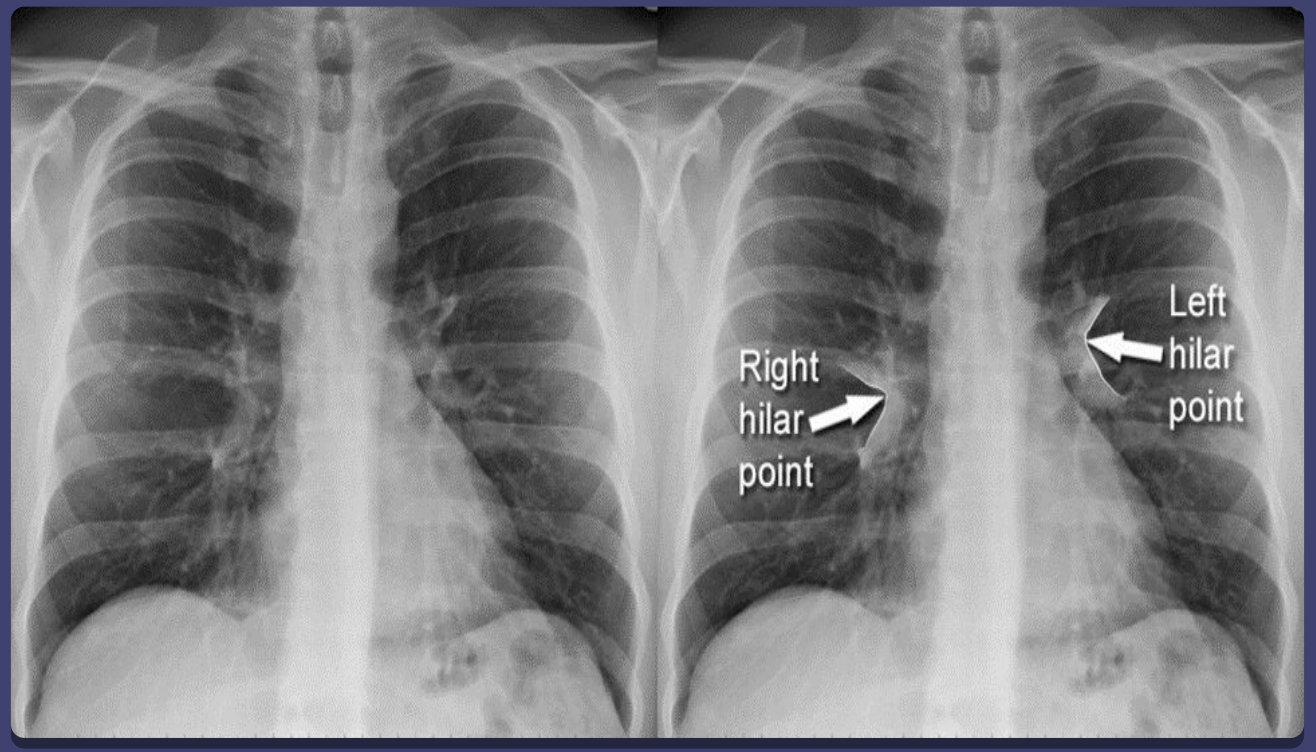


Hilar enlargement

The hilum contains pulmonary artery and vein, lymph nodes and bronchi. So any pathology in this region usually originate from one of these components.

The criteria of a normal hilum:

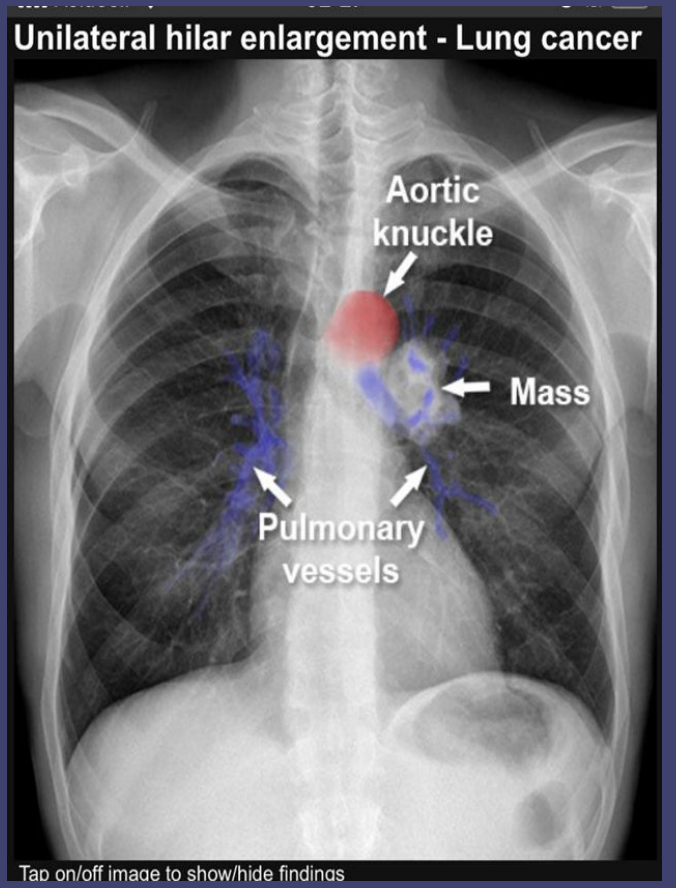
- Location: LT should be higher than RT
- Shape: crescent or C shaped
- Size or bulk : bilaterally symmetrical in size.



Unilateral Hilar Mass

1. **Neoplasms including:**
   - Lung cancer: most common cause.
   - Metastasis
   - Lymphoma
2. **Inflammation : especially Tuberculosis.**





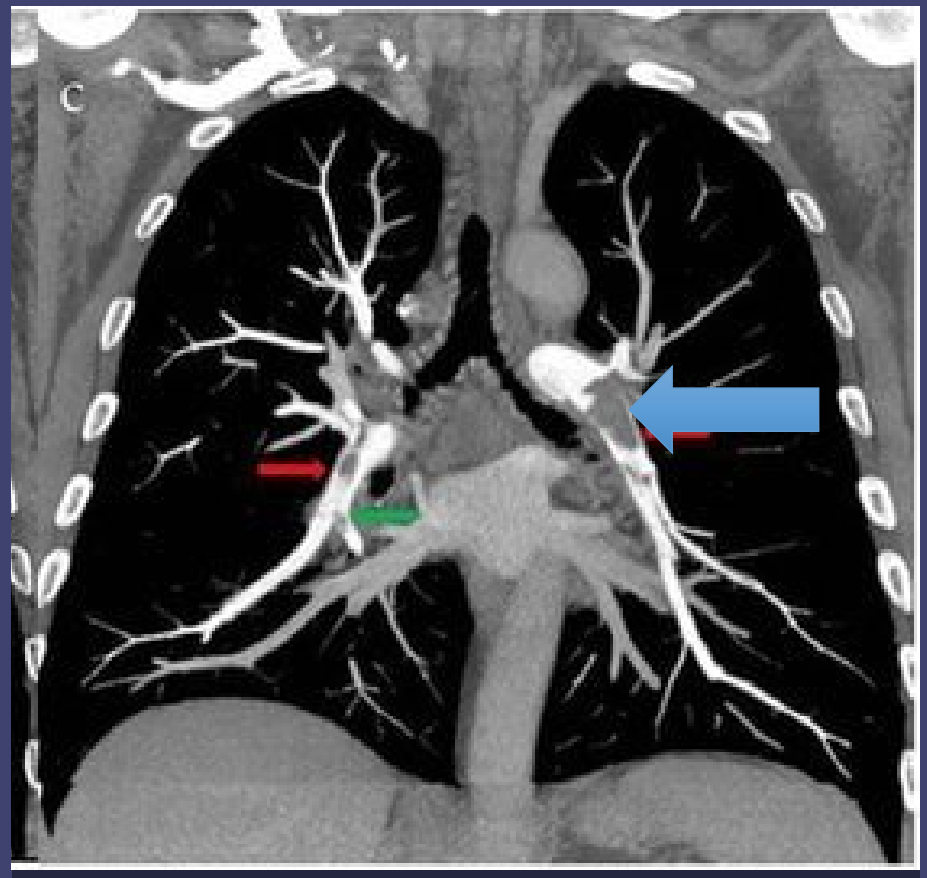
Bilateral hilar masses

- **A.<u>Neoplasm</u>**
  - Lymphoma/Leukemia
  - Metastases
- **<u>B. Inflammation</u>**
  - **Sarcoidosis** :
- **C. <u>Large pulmonary arteries</u>:** pulmonary hypertension,.



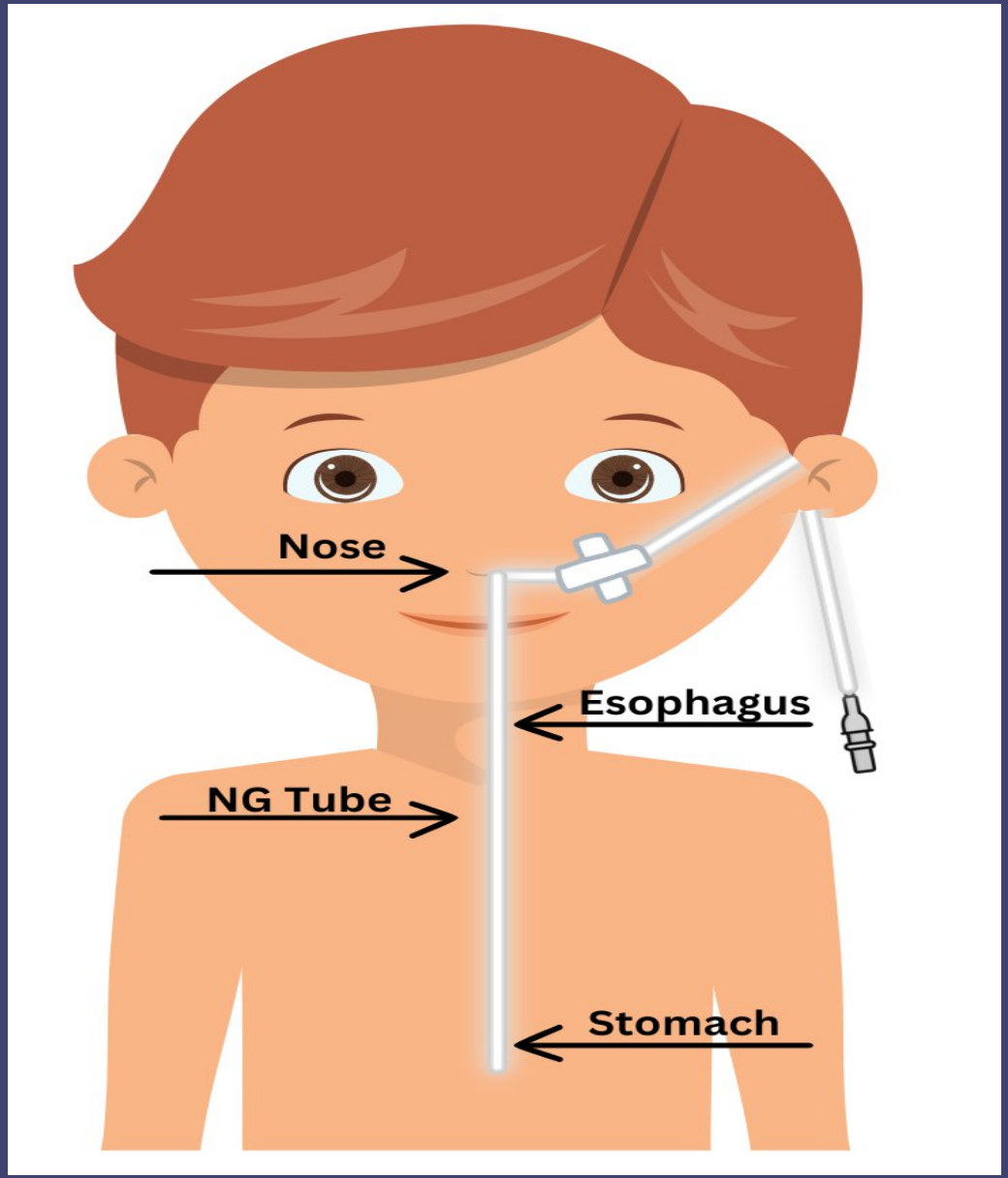
Imaging of Pulmonary Arteries

- CXR is usually normal in acute Pulmonary Embolism (PE).
- **CT pulmonary angiogram CTPA is the modality of choice when PE has to be confirmed/excluded.**
- CTPA shows filling defects within the pulmonary arteries indicating embolism.



CT angiogram showing filling defects (emboli) within the pulmonary arteries.

Tubes and Devices on CXR



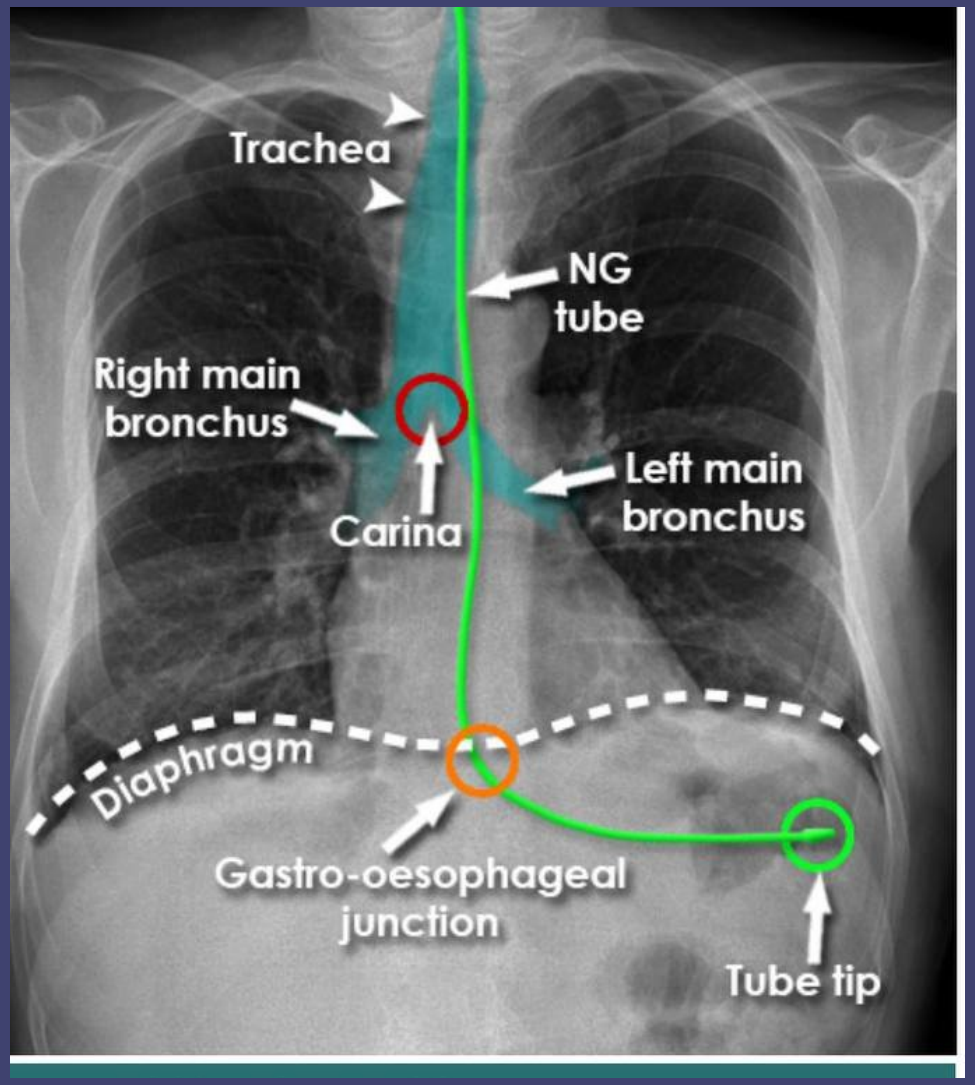
NG Tube

In the MIDLINE

Tip below diaphragm

Bisect the carina

Tip is 10 cm from GO junction

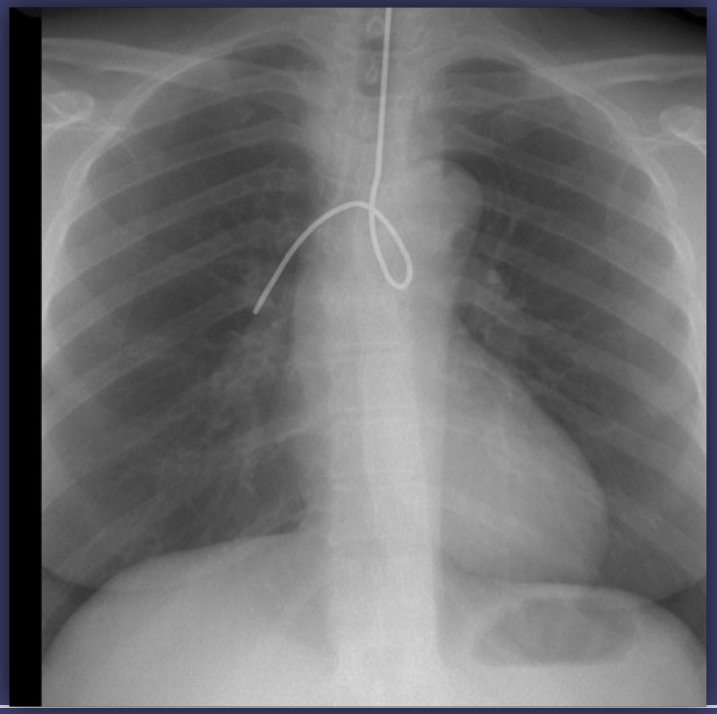


NG Tube Mispositioning

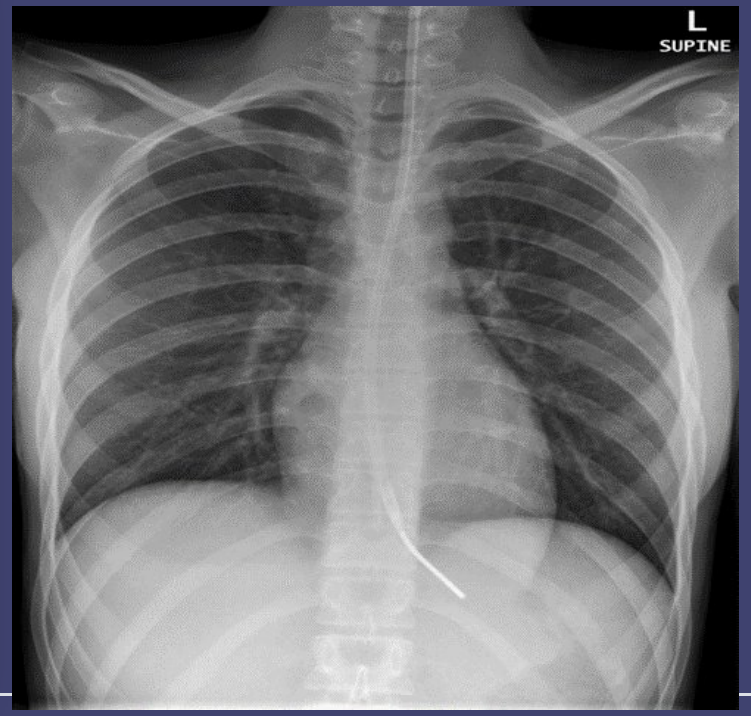
In the LT bronchus



Coiled within esophagus

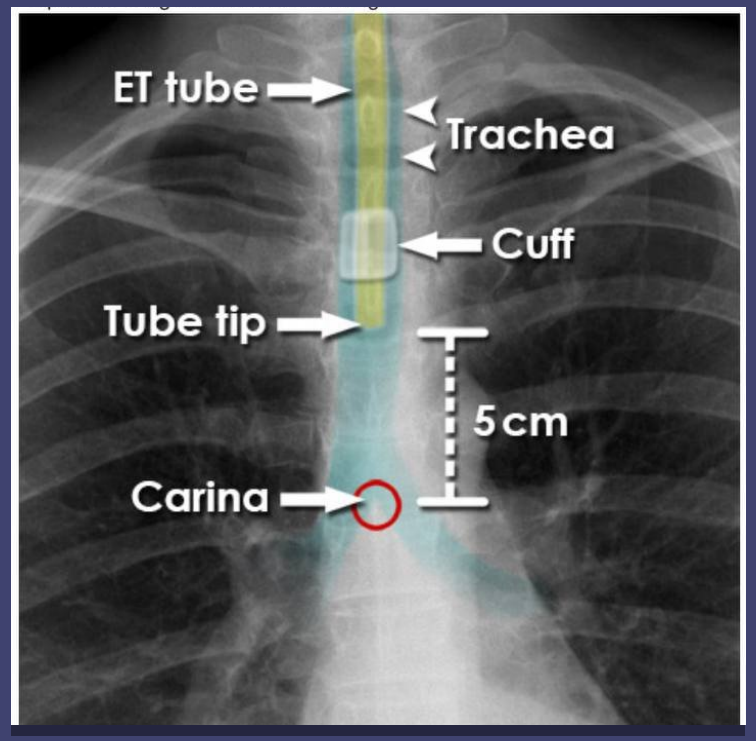


Short (risk of aspiration)

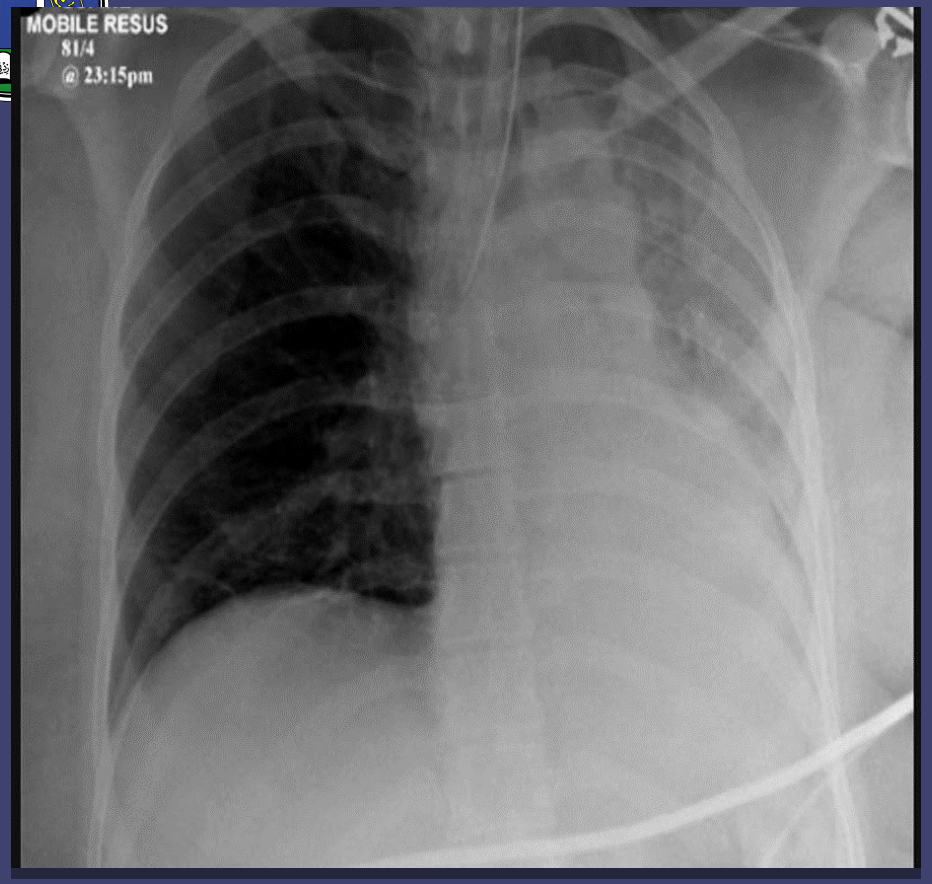


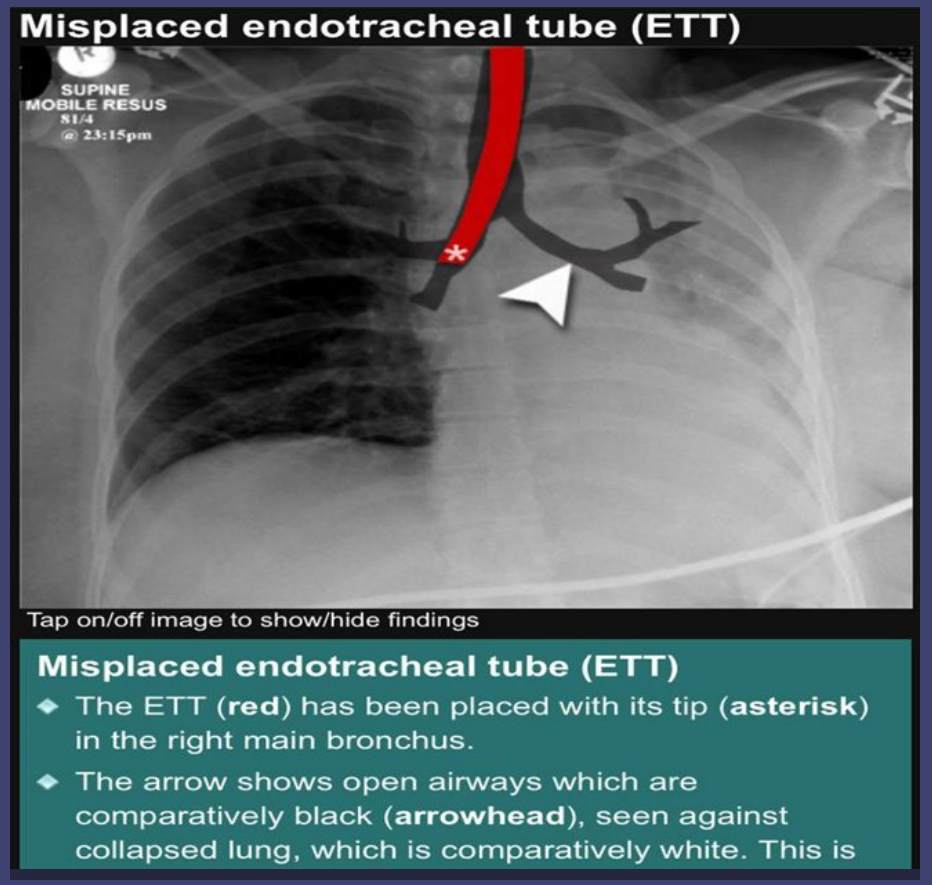
ENDOTRACHEAL TUBE

- The tip should be seen 5 cm above the carina.
- A misplaced tube that pass into the main bronchus of one side will lead to lung collapse of the other side.





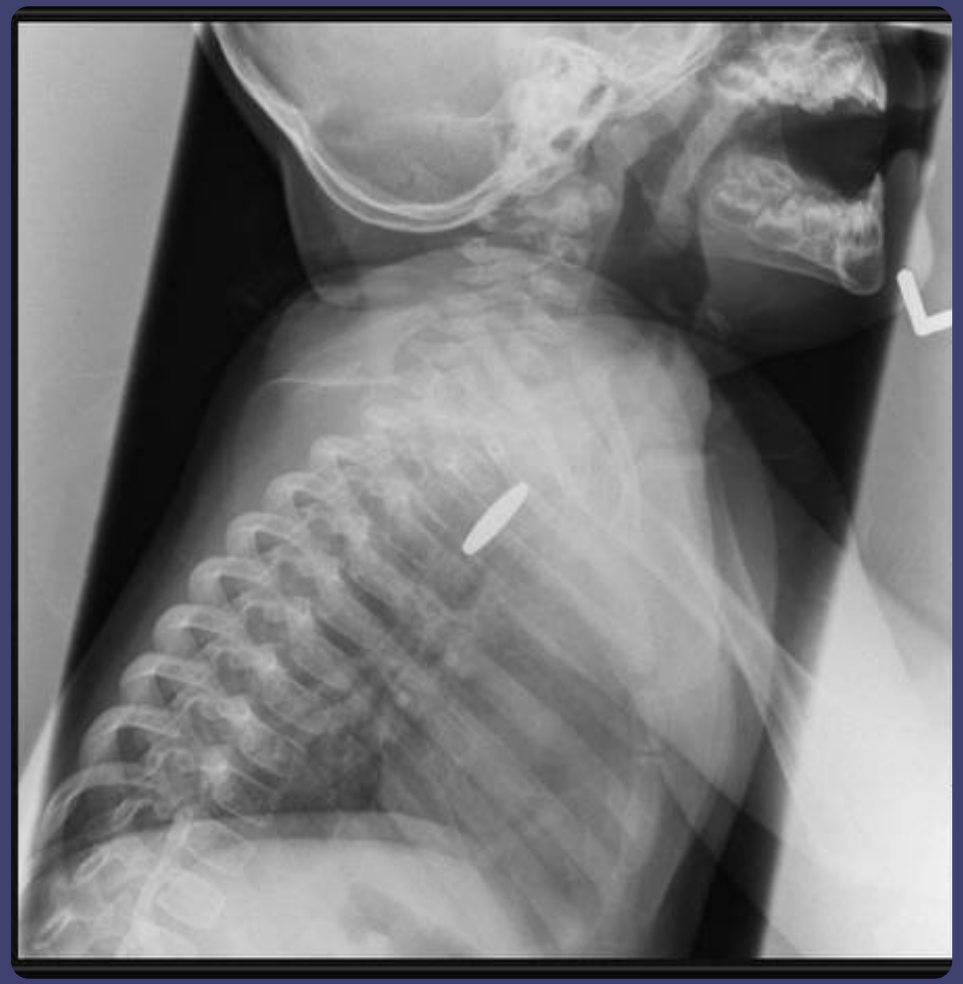


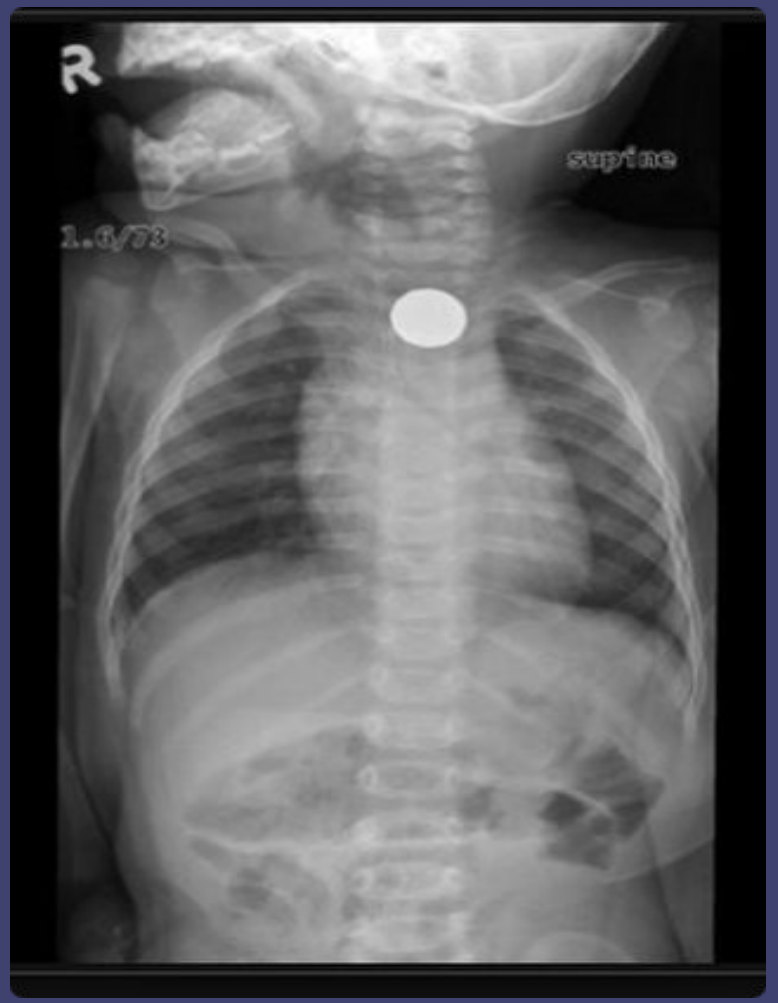


ET Tube passing into the right main bronchus leading to total lung collapse on the left

Foreign Bodies on CXR

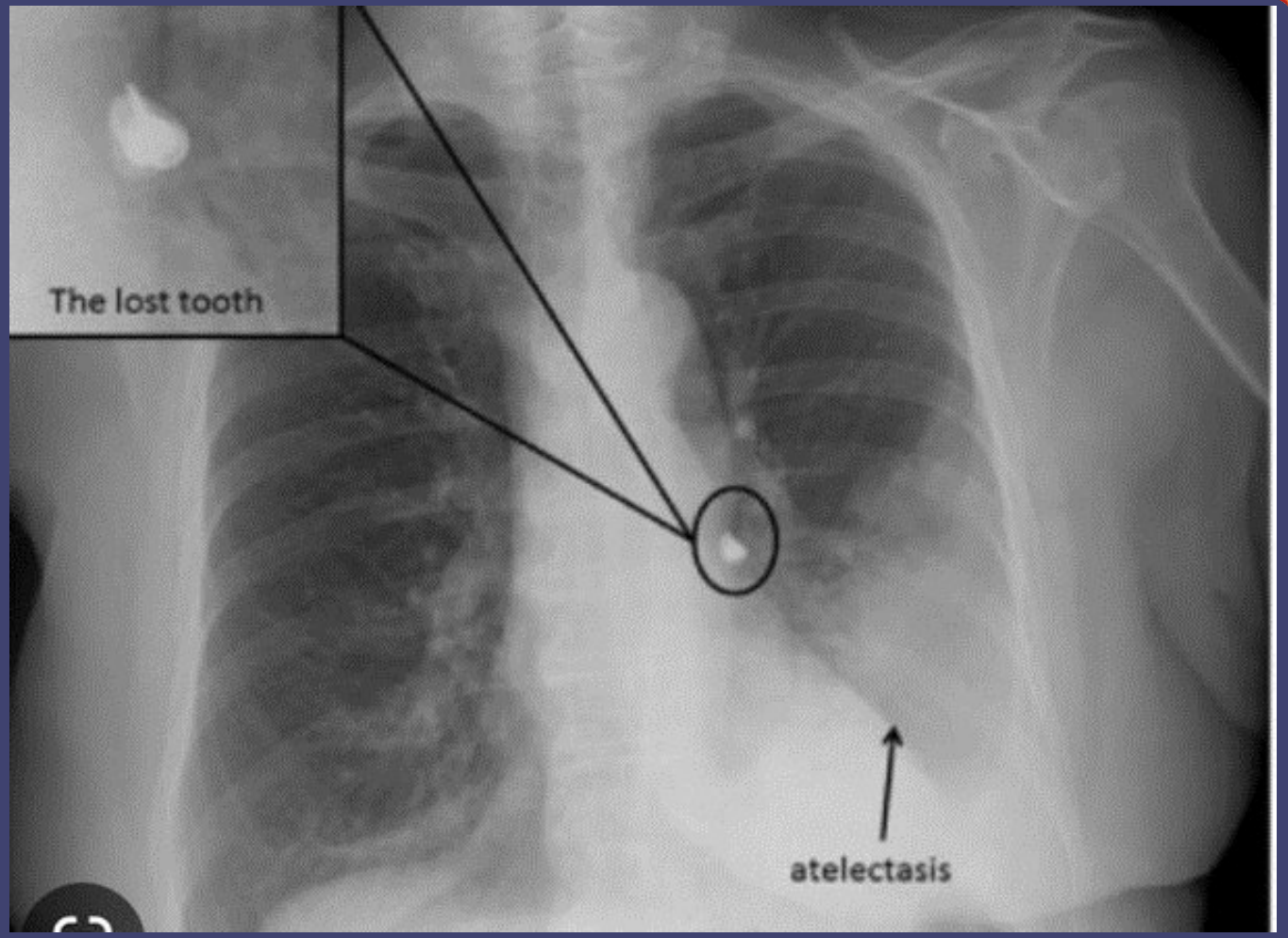
- Inhalation/ingestion of foreign bodies is a common emergency in pediatrics .
- CXR (PA and Lateral views) should be performed for evaluation.
- (PA view alone is insufficient).
- Any artifact related to the patient's skin/clothes should be seen external to the lung on the lateral view.





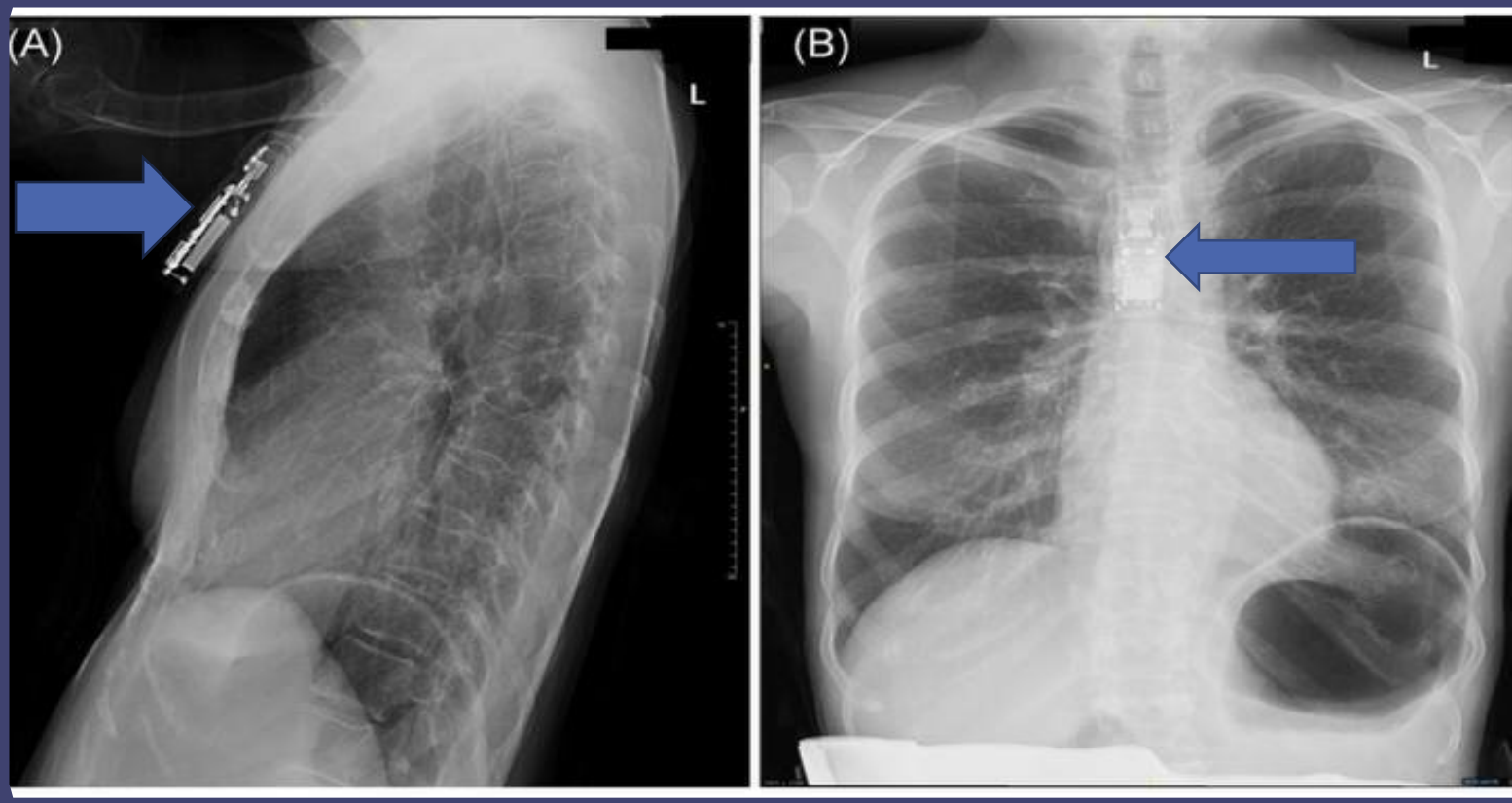
Foreign body in the esophagus

Accidentally ingested tooth (common scenario in elderly people). The tooth can be seen as radiopaque FB at the level of LT bronchus



Metal Objects Artefacts

ANY METAL OBJECT RELATED TO THE SKIN/ CLOTHES SHOULD APPEAR EXTERNAL TO THE LUNG ON LATERAL VIEW.



Summary

- CT is the modality of choice for assessment of mediastinal pathologies including acute aortic syndrome
- Pulmonary embolism is one of the disease that appear normal on CXR. CTPA is the modality of choice.
- Hilar pathologies are always serious and can be easily missed. Adopt a well structured system for evaluation on CXR.
- Misplacement of NG and ET tubes are not uncommon. Make sure you know how to assess on CXR
- Don’t ever miss a FB on CXR

References

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2. Armstrong P, Wastie M, Rockall AG. Diagnostic Imaging. John Wiley & Sons; 2010

QUESTIONS & DISCUSSION

Thank you

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